

2010
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15, 17, 19, 21 and 23

Total marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–31
- Allow about 1 hour and 40 minutes for this part

Section II Pages 25–36

25 marks

- Attempt ONE question from Questions 32–36
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

1 Water is released during a polymerisation reaction.

Which monomer is likely to have been involved in the reaction?

- (A) Ethene
- (B) Glucose
- (C) Styrene
- (D) Vinyl chloride

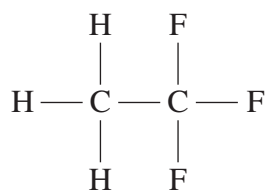
2 Which of the following is an example of a transuranic element?

- (A) C-14
- (B) Co-60
- (C) U-238
- (D) Cm-249

3 Which substance shows the correct indicator colour?

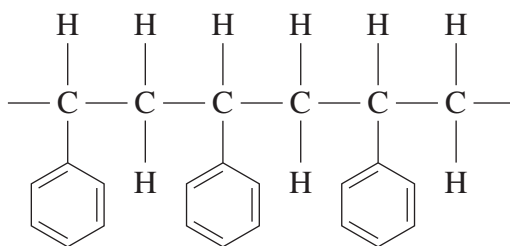
	<i>Substance</i>	<i>pH</i>	<i>Indicator</i>	<i>Colour</i>
(A)	Stomach acid	2	Methyl orange	Yellow
(B)	Lemon juice	3	Phenolphthalein	Pink
(C)	Soda water	4	Phenolphthalein	Pink
(D)	Seawater	8	Methyl orange	Yellow

- 4 The diagram shows the structural formula of a gas.



How many isomers does this compound have?

- (A) 1
(B) 2
(C) 3
(D) 4
- 5 An imbalance of which two substances causes the eutrophication of waterways?
- (A) H^+ and OH^-
(B) Mg^{2+} and Ca^{2+}
(C) Oxygen and ozone
(D) Phosphorus and nitrogen
- 6 The diagram shows a section of a polymer.



What is the systematic name of the monomer?

- (A) Polybenzene
(B) Benzylethene
(C) Ethylbenzene
(D) Ethenylbenzene

- 7 Equal volumes of four 0.1 mol L^{-1} acids were titrated with the same sodium hydroxide solution.

Which one requires the greatest volume of base to change the colour of the indicator?

- (A) Citric acid
- (B) Acetic acid
- (C) Sulfuric acid
- (D) Hydrochloric acid

- 8 In a research report a student wrote, 'Acids are compounds that contain hydrogen and can dissolve in water to release hydrogen ions into solution.'

Who originally stated this theory of acids?

- (A) Arrhenius
- (B) Brönsted–Lowry
- (C) Davy
- (D) Lavoisier

- 9 What types of reaction occur in the Haber process during the production of ammonia?

- (A) Redox and synthesis
- (B) Hydration and redox
- (C) Decomposition and oxidation
- (D) Reduction and decomposition

- 10 A sample of water from a stream, suspected to be contaminated with metal ions, was analysed.

The results of some tests on the water are recorded in the table.

<i>Test</i>	<i>Result</i>
Add dilute HCl	No change
Add Na ₂ SO ₄ solution	White precipitate formed
Flame test	Pale green colour

What is the most likely contaminant in the water?

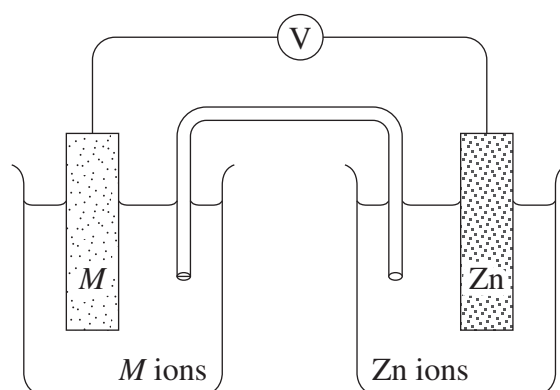
- (A) Ba²⁺
 - (B) Ca²⁺
 - (C) Cu²⁺
 - (D) Fe³⁺
- 11 An organic liquid, when reacted with concentrated sulfuric acid, produces a compound that decolourises bromine water.

What is the formula of the organic liquid?

- (A) C₆H₁₂
 - (B) C₆H₁₄
 - (C) C₆H₁₁OH
 - (D) C₅H₁₁COOH
- 12 In which of the following reactions does the metal atom show the greatest change in oxidation state?

- (A) MnO₄⁻ to Mn²⁺
- (B) MnO₂ to Mn(OH)₃
- (C) PbO₂ to PbSO₄
- (D) VO₂⁺ to VO²⁺

- 13 The diagram shows a galvanic cell.



Which of the following metals (*M*) acting as an anode would produce the lowest theoretical potential for the cell?

- (A) Calcium
 - (B) Copper
 - (C) Iron
 - (D) Manganese
- 14 The table shows information about three carbon compounds.

<i>Compound</i>	<i>Structural formula</i>	<i>Molecular weight</i>	<i>Boiling point</i>
<i>X</i>	$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{O} - \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array} $	60	97°C
<i>Y</i>	$ \begin{array}{ccc} & \text{H} & \\ & & \\ \text{H} & - \text{C} & - \text{C} \begin{array}{l} \nearrow \text{O} \\ \searrow \text{O} - \text{H} \end{array} \\ & & \\ & \text{H} & \end{array} $	60	118°C
<i>Z</i>	$ \begin{array}{ccc} & \text{H} & & \text{O} \\ & & & \\ \text{H} & - \text{C} & - \text{O} & - \text{C} - \text{H} \\ & & & \\ & \text{H} & & \end{array} $	60	?

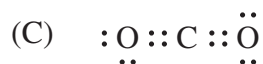
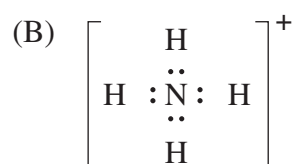
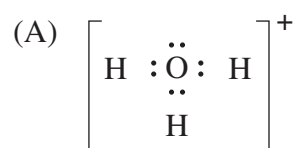
What is the best estimate for the boiling point of compound *Z*?

- (A) 31°C
- (B) 101°C
- (C) 114°C
- (D) 156°C

15 What mass of ethanol is obtained when 5.68 g of carbon dioxide is produced during fermentation, at 25°C and 100 kPa?

- (A) 2.95 g
- (B) 5.95 g
- (C) 33.6 g
- (D) 147.2 g

16 Which of the following Lewis structures does NOT contain a coordinate covalent bond?



17 A student completed an experiment to determine the amount of energy absorbed by a volume of water.

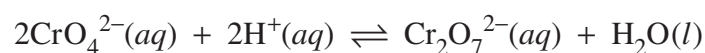
The following data were recorded.

Mass of beaker	215.6 g
Mass of beaker plus water	336.1 g
Final temperature of water	71.0°C
Energy absorbed	21.2 kJ

What was the initial temperature of the water?

- (A) 15°C
- (B) 25°C
- (C) 29°C
- (D) 42°C

- 18 Chromate and dichromate ions form an equilibrium according to the following equation.



Which solution would increase the concentration of the chromate ion (CrO_4^{2-}) when added to the equilibrium mixture?

- (A) Sodium nitrate
 - (B) Sodium chloride
 - (C) Sodium acetate
 - (D) Ammonium chloride
- 19 Sodium azide is used in automobile airbags to provide a source of nitrogen gas for rapid inflation in an accident. The equation shows the production of nitrogen gas from sodium azide.



What mass of sodium azide will produce 40 L of N_2 at 100 kPa and 0°C ?

- (A) 70 g
 - (B) 76 g
 - (C) 114 g
 - (D) 172 g
- 20 Solutions containing copper ions were analysed by AAS. A standard solution of 10 ppm copper had an AAS absorbance of 0.400. A second solution of unknown concentration was found to have an absorbance of 0.500.

100 mL of this second solution was reacted with excess sodium carbonate solution. The precipitate was then dried and weighed.

What mass of precipitate was formed?

- (A) 1.25×10^{-3} g
- (B) 2.43×10^{-3} g
- (C) 1.54 g
- (D) 2.43 g

--	--	--	--	--

Centre Number

Section I (continued)

--	--	--	--	--	--	--	--	--

Student Number

Part B – 55 marks

Attempt Questions 21–31

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Show all relevant working in questions involving calculations.

Question 21 (3 marks)

A 0.001 mol L^{-1} solution of hydrochloric acid and a 0.056 mol L^{-1} solution of ethanoic acid both have a pH of 3.0. 3

Why do both solutions have the same pH?

.....

.....

.....

.....

.....

.....

Question 22 (6 marks)

A student prepared the compound methyl propanoate in a school laboratory.

- (a) Give a common use for the class of compounds to which methyl propanoate belongs. **1**

.....
.....

- (b) In the preparation of this compound a few drops of concentrated sulfuric acid were added to the starting materials. The mixture was then refluxed for a period of time. **2**

Why was it necessary to reflux the mixture?

.....
.....
.....
.....

- (c) Name the TWO reactants used in preparing the methyl propanoate and draw their structural formulae. **3**

.....
.....
.....
.....
.....
.....

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 23 (3 marks)

- (a) Write a balanced chemical equation for the complete combustion of 1-butanol. 1

.....

.....

- (b) A student measured the heat of combustion of three different fuels. The results are shown in the table. 2

<i>Fuel</i>	<i>Heat of combustion (kJ g⁻¹)</i>
<i>A</i>	–48
<i>B</i>	–38
<i>C</i>	–28

The published value for the heat of combustion of 1-butanol is 2676 kJ mol⁻¹.

Which fuel from the table is likely to be 1-butanol? Justify your answer.

.....

.....

.....

.....

.....

.....

Question 24 (4 marks)

In the margarine industry, alkenes are often hydrogenated to convert unsaturated oils into solid fats that have a greater proportion of saturated molecules.

- (a) Using ethene as an example, write an equation for this reaction and state the type of reaction this represents. **2**

.....

.....

.....

- (b) Describe a test that could be used to confirm that all the ethene has been converted. **2**

.....

.....

.....

.....

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 25 (5 marks)

What is the relationship between dissolved oxygen and biochemical oxygen demand and why is it important to monitor both in natural waterways?

5

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

BLANK PAGE

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 26 (4 marks)

A gas is produced when 10.0 g of zinc is placed in 0.50 L of 0.20 mol L⁻¹ nitric acid.

4

Calculate the volume of gas produced at 25°C and 100 kPa. Include a balanced chemical equation in your answer.

.....

.....

.....

.....

.....

.....

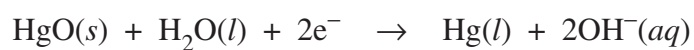
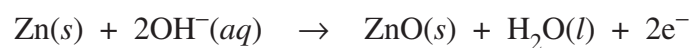
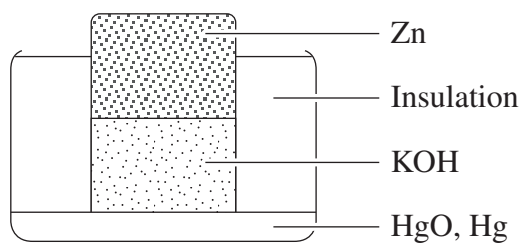
.....

.....

Question 27 (2 marks)

The diagram shows a particular cell with relevant half equations.

2



Identify the anode, cathode and electrolyte for this cell.

.....

.....

.....

.....

--	--	--	--	--

Centre Number

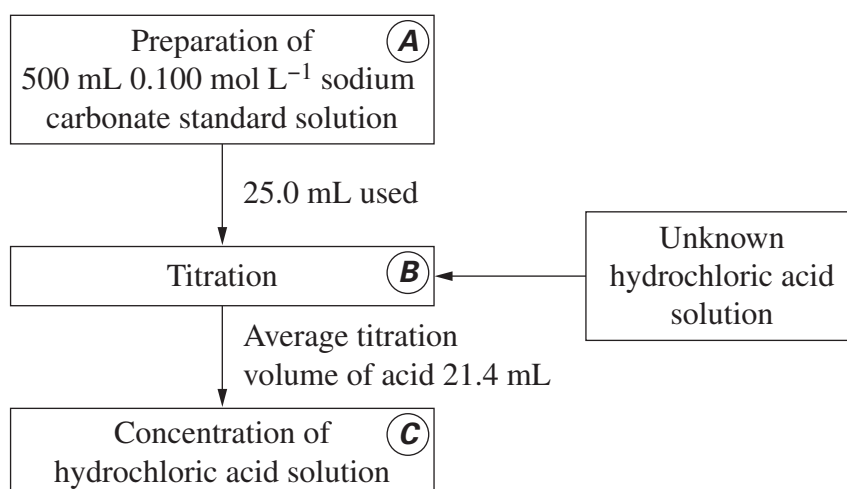
Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 28 (8 marks)

The flowchart shown outlines the sequence of steps used to determine the concentration of an unknown hydrochloric acid solution. 8



Describe steps **A**, **B** and **C** including correct techniques, equipment and appropriate calculations. Determine the concentration of the hydrochloric acid.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 28 continues on page 18

Question 28 (continued)

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

End of Question 28

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

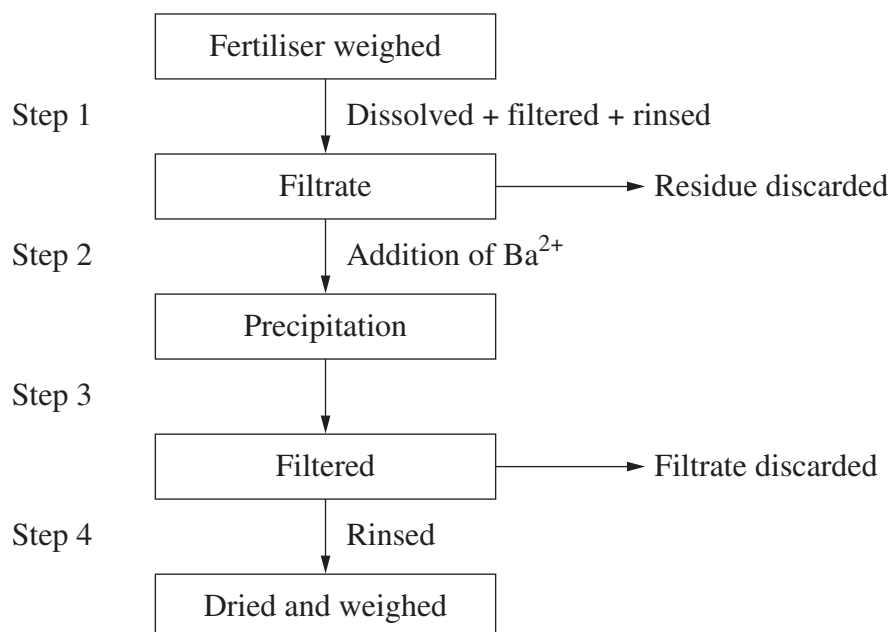
Student Number

Question 29 (6 marks)

Please turn over

Question 29 (6 marks)

The flowchart shown outlines the process used to determine the amount of sulfate present in a sample of lawn fertiliser.



- (a) What assumptions were made and how do these affect the validity of this process? **3**

.....

.....

.....

.....

.....

.....

- (b) It was found that 4.25 g had a sulfate content of 35%. **3**

What is the mass of the dried precipitate at Step 4? Include a chemical equation in your answer.

.....

.....

.....

.....

.....

.....

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 30 (8 marks)

- (a) Compare the process of polymerisation of ethylene and glucose. Include relevant chemical equations in your answer. 3

.....

.....

.....

.....

.....

.....

.....

Question 30 continues on page 22

Question 30 (continued)

- (b) Explain the relationship between the structures and properties of THREE different polymers from ethylene and glucose, and their uses. **5**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

End of Question 30

--	--	--	--	--

Centre Number

Section I – Part B (continued)

--	--	--	--	--	--	--	--	--

Student Number

Question 31 (6 marks)

- (a) A student collected a 250 mL sample of water from a local dam for analysis. The data collected are shown in the table.

Mass of filter paper	0.23 g
Mass of filter paper and solid	0.47 g
Mass of evaporating basin	43.53 g
Mass of basin and solid remaining	44.67 g

- (i) The water was filtered and the filtrate evaporated to dryness. 2

Calculate the percentage of the total dissolved solids in the dam sample.

.....

.....

.....

.....

- (ii) It is suspected that the water in the dam has a high concentration of chloride ions. 2

Describe a chemical test that could be carried out on the water sample to determine the presence of chloride ions. Include an equation in your answer.

.....

.....

.....

.....

Question 31 continues on page 24

Question 31 (continued)

- (b) Name an ion other than chloride that commonly pollutes waterways, and identify its source and the effect of its presence on water quality. **2**

.....

.....

.....

.....

End of Question 31

Chemistry

Section II

25 marks

Attempt ONE question from Questions 32–36

Allow about 45 minutes for this section

Answer parts (a)–(c) of the question in a writing booklet. Answer parts (d)–(e) of the question in a SEPARATE writing booklet.

Show all relevant working in questions involving calculations.

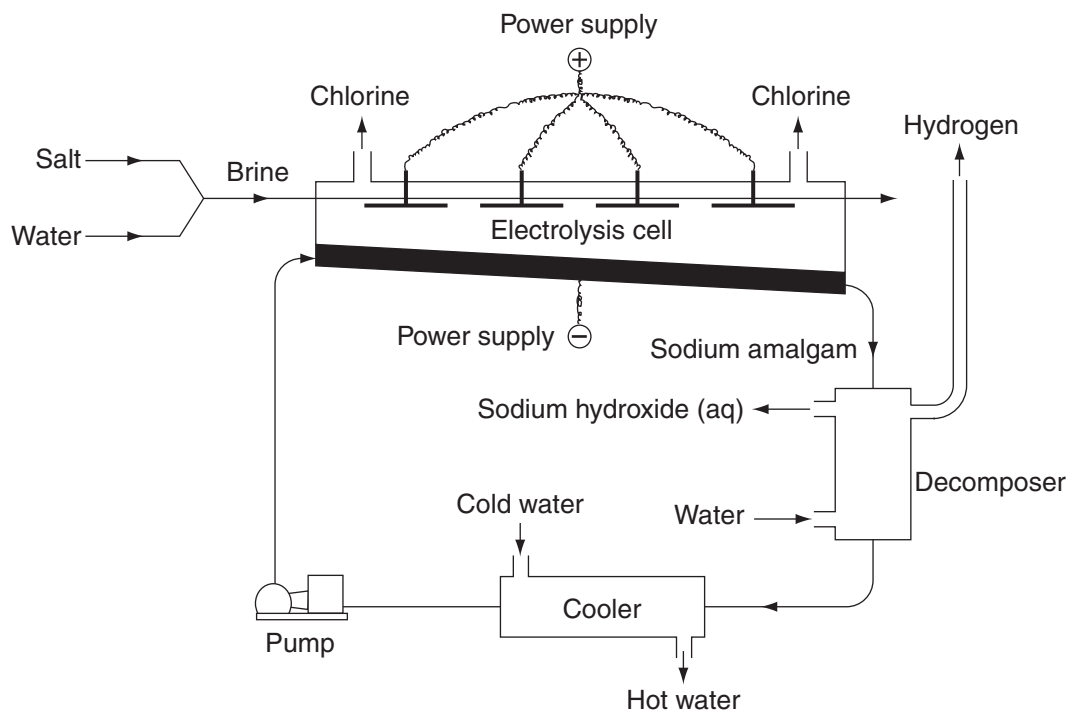
	Pages
Question 32 Industrial Chemistry	26–27
Question 33 Shipwrecks, Corrosion and Conservation	28–29
Question 34 The Biochemistry of Movement	30–31
Question 35 The Chemistry of Art	32–33
Question 36 Forensic Chemistry	34–36

Question 32 — Industrial Chemistry (25 marks)

Answer parts (a)–(c) in a writing booklet.

- (a) Identify the type of cell shown and outline the process used in the extraction of sodium hydroxide.

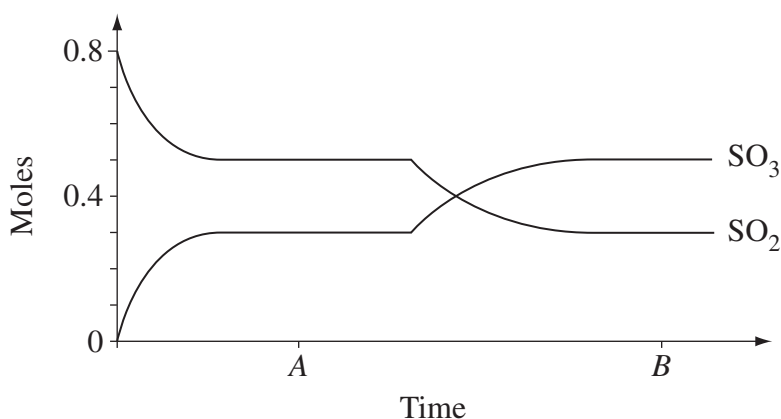
3



- (b) Compare the electrolysis of molten sodium chloride and aqueous sodium chloride. Write the relevant half equations and overall reaction for each process.

5

- (c) At room temperature 0.80 moles of SO_2 and 0.40 moles of O_2 were introduced into a sealed 10 L vessel and allowed to come to equilibrium.



- (i) Write the equilibrium constant expression and calculate the value for the equilibrium constant at time A.
- (ii) Explain why a new equilibrium position was established at time B.

3

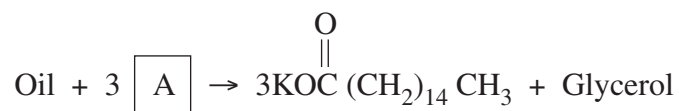
2

Question 32 continues on page 27

Question 32 (continued)

Answer parts (d)–(e) in a SEPARATE writing booklet.

(d) The equation represents a reaction that can be performed in a school laboratory.



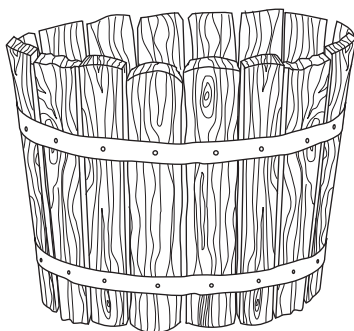
- (i) Identify both this type of reaction and the reactant A. **2**
- (ii) Describe how this type of reaction could be carried out in a school laboratory including specific safety precautions for this process. **3**
- (e) Assess both the importance and resulting environmental impacts of using limestone in the Solvay Process. **7**

End of Question 32

Question 33 — Shipwrecks, Corrosion and Conservation (25 marks)

Answer parts (a)–(c) in a writing booklet.

- (a) The following artefact was retrieved from a ship that sank 150 years ago off the coast of New South Wales. **3**



Outline the effect that the marine environment would have had on the artefact.

- (b) (i) Use a fully labelled diagram to show the electrolysis of an aqueous solution of potassium chloride. Write the relevant half equations and the overall reaction for the cell. **4**
- (ii) How would the cathode be identified? **1**
- (c) The following table shows the composition of four types of steel. **5**

<i>Steel</i>	<i>Composition</i>
1	99.8% Fe, 0.2% C
2	98.5% Fe, 1.5% C
3	94% Fe, 4% C, 1% Mn, 1% Si
4	75% Fe, 15% Cr, 10% Ni

Explain how the composition of each type of steel determines its properties and uses.

Question 33 continues on page 29

Question 33 (continued)

Answer parts (d)–(e) in a SEPARATE writing booklet.

- | | | | |
|-----|------|--|----------|
| (d) | (i) | An investigation into environmental factors that affect the rate of corrosion of iron can be performed in a school laboratory. | 4 |
| | | Describe how you could perform this investigation in relation to THREE environmental factors. | |
| | (ii) | Explain how the effect of ONE of the factors could be reduced in a marine environment. | 1 |
| (e) | | Evaluate the suitability of techniques used for restoring and conserving wooden and copper artefacts that have been immersed in salt water for at least 100 years. | 7 |

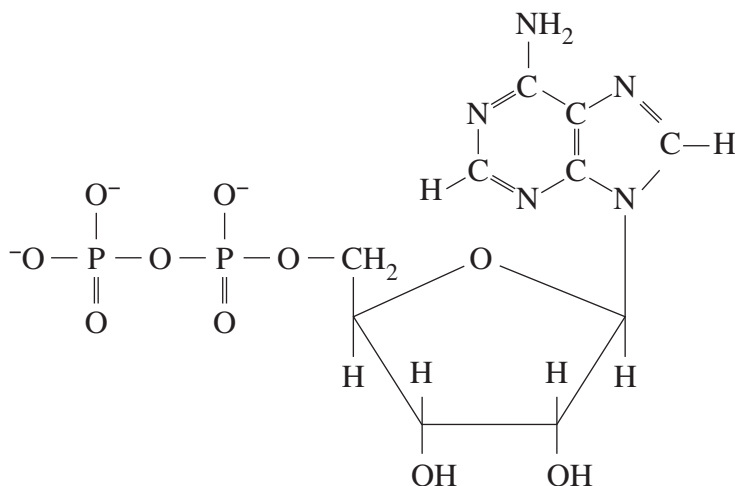
End of Question 33

Question 34 — The Biochemistry of Movement (25 marks)

Answer parts (a)–(c) in a writing booklet.

- (a) Identify this molecule and outline its importance in cellular metabolism.

3



- (b) (i) If the oxidation of glucose in the body occurred as rapidly as when glucose is combusted in air, the sudden release of this large amount of energy could not be utilised by cells.

2

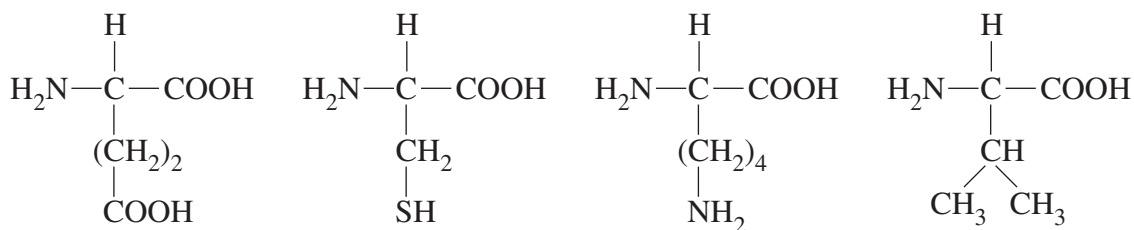
How do cells overcome this problem?

- (ii) Explain the change in cellular pH during anaerobic respiration. Include a balanced chemical equation in your answer.

3

- (c) Explain in terms of chemical bonding how the secondary and tertiary structures of a protein depend on the amino acids in its primary structure. In your answer, make specific reference to the structures shown.

5



Question 34 continues on page 31

Question 34 (continued)

Answer parts (d)–(e) in a SEPARATE writing booklet.

- (d) Construct a flowchart that summarises the steps in aerobic respiration and account for the total energy output. **5**

- (e) Analyse how an understanding of the structure and function of skeletal muscle cells has influenced training programs for athletes. **7**

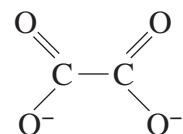
End of Question 34

Question 35 — The Chemistry of Art (25 marks)

Answer parts (a)–(c) in a writing booklet.

- (a) Identify the element in period 3 of the periodic table that has the highest electronegativity and justify your choice. **3**

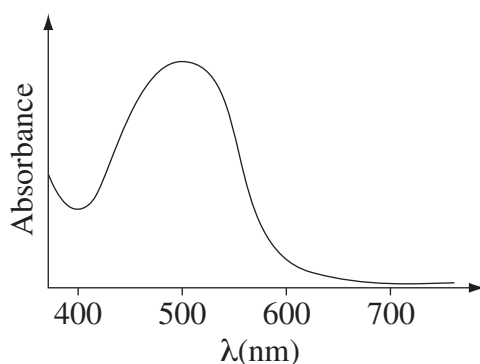
- (b) (i) What type of ligand is represented by the oxalate ion shown? **1**



- (ii) The oxalate ion forms complex ions with both Cu^{2+} and Cu^+ . **4**

State whether each of these complex ions is coloured or not and justify your answer.

- (c) (i) The absorbance spectrum of a commercial pigment is shown. **2**



Draw the resulting reflectance spectrum for this pigment and predict its colour.

- (ii) Outline the use of infra-red radiation in both the analysis and identification of pigments. **3**

Question 35 continues on page 33

Question 35 (continued)

Answer parts (d)–(e) in a SEPARATE writing booklet.

- (d) Experimental evidence from emission line spectra of gaseous atoms has highlighted both the merits and the limitations of Bohr’s atomic model. **5**

Discuss Bohr’s atomic model with reference to this evidence.

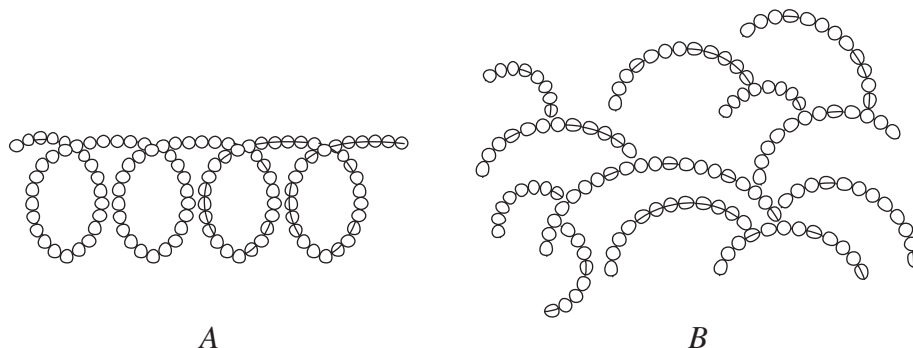
- (e) Evaluate the potential health risks associated with the use of a range of cosmetics by ancient cultures. **7**

End of Question 35

Question 36 — Forensic Chemistry (25 marks)

Answer parts (a)–(c) in a writing booklet.

- (a) The diagrams show the arrangement of glucose units in two polysaccharides. **3**



○ Single glucose unit

Identify *A* and *B* and outline their differences in structure and origin.

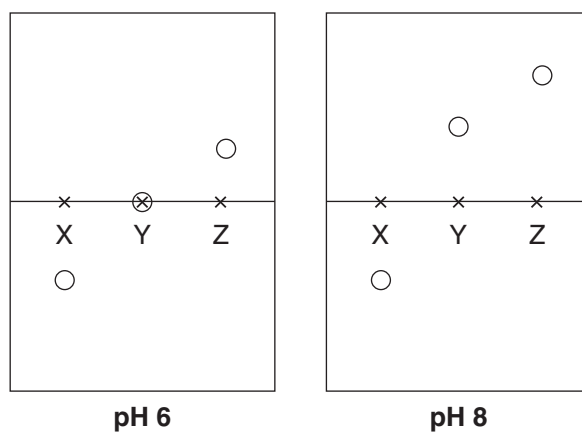
Question 36 continues on page 35

Question 36 (continued)

(b) The structures of three amino acids at pH 6 and pH 8 are shown.

	Glutamic acid	Lysine	Valine
pH 6	$ \begin{array}{c} \text{O}=\text{C}-\text{O}^- \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{N}^+-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{N}-\text{H}^+ \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{N}^+-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ \\ \text{H}-\text{N}^+-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $
pH 8	$ \begin{array}{c} \text{O}=\text{C}-\text{O}^- \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{N}-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{N}-\text{H}^+ \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{C}-\text{H} \\ \\ \text{H}-\text{N}-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $	$ \begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \quad \text{H} \\ \quad \quad \quad \\ \text{H}-\text{C}-\text{C}-\text{C}-\text{H} \\ \quad \quad \\ \text{H} \quad \text{H} \quad \text{H} \\ \\ \text{H}-\text{N}-\text{C}-\text{C}(=\text{O})\text{O}^- \\ \quad \\ \text{H} \quad \text{H} \end{array} $

Samples of each amino acid underwent electrophoresis at pH 6 and pH 8. The results are shown below.

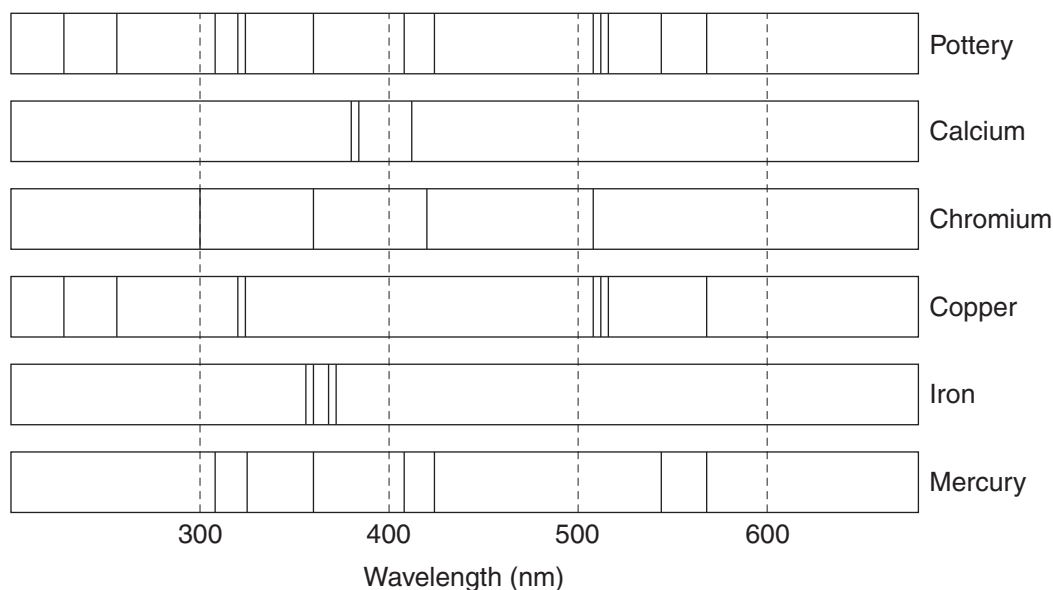


- (i) Outline the process of electrophoresis used in identifying amino acids. 2
- (ii) Identify the amino acids **X**, **Y** and **Z** and justify your answer. 3

Question 36 continues on page 36

Question 36 (continued)

- (c) The following emission spectra were used to identify the manufacturer of a piece of pottery.



<i>Manufacturer</i>	<i>Metal atoms in pottery</i>
A	copper, chromium
B	copper, mercury
C	calcium, chromium

- (i) Which manufacturer made the pottery – A, B or C? 1
- (ii) Explain the production and use of emission spectra in identifying the manufacturer of this pottery. 4
- Answer parts (d)–(e) in a SEPARATE writing booklet.
- (d) A forensic chemist analyses soil from a pair of shoes worn by a person suspected of committing a crime.
- (i) Identify FOUR properties of soil a forensic chemist would investigate. 2
- (ii) Describe both an organic and an inorganic test that could be performed on the soil that could match it to soil at the crime scene. 3
- (e) Describe the techniques used to analyse DNA and the applications of these techniques in forensic analysis. 7

End of paper

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

Awaiting copyright



2010 HSC Chemistry Marking Guidelines

Section I, Part A

Question	Answer
1	B
2	D
3	D
4	B
5	D
6	D
7	A
8	A
9	A
10	A
11	C
12	A
13	D
14	A
15	B
16	C
17	C
18	C
19	B
20	B

Section I, Part B

Question 21

Criteria	Marks
- Identifies HCl as a strong acid which fully ionises in aqueous solution - Identifies ethanoic acid as a weak acid which does not fully ionise in aqueous solution - Identifies that the total concentration of ethanoic acid must be higher than that of HCl to give an equivalent $[H^+]$ and therefore pH, as $pH = -\log_{10} [H^+]$	3
- Identifies HCl as a strong acid or fully ionised - Identifies CH_3COOH acid as a weak acid or partially ionised - Identifies that CH_3COOH concentration needs to be higher than HCl to give equivalent $[H^+]$ and therefore pH	2
H^+ concentration from HCl is equal to H^+ from CH_3COOH OR - Identifies HCl as a strong acid or is fully ionised and identifies CH_3COOH as a weak acid or is partially ionised	1

Question 22 (a)

Criteria	Marks
Gives a valid use of esters	1

Question 22 (b)

Criteria	Marks
• Gives valid reasons for refluxing the mixture	2
• Gives a valid reason for refluxing the mixture	1

Question 22 (c)

Criteria	Marks
• Draws correct structural formulae and correctly names both molecules	3
• Draws correct structural formulae for both molecules and correctly names ONE OR • Draws correct structural formula for ONE molecule and correctly names both	2
• Gives correct names for both molecules OR • Draws correct structural formula for ONE	1

Question 23 (a)

Criteria	Marks
• Gives correctly balanced equation	1

Question 23 (b)

Criteria	Marks
• Calculates kJ mol^{-1} values correctly for each fuel or kJ g^{-1} for 1-butanol • Gives correct reason for identifying correct fuel	2
• Identifies correct fuel with all calculations shown but no reason OR • Identifies correct fuel with correct reasons but incorrect calculations OR • Identifies incorrect fuel but has correct calculations	1

Question 24 (a)

Criteria	Marks
• Gives balanced chemical equation and identifies type of reaction	2
• Gives balanced chemical equation OR • Identifies type of reaction	1

Question 24 (b)

Criteria	Marks
- States that bromine water decolourises in the presence of ethene - States that on adding bromine water sample should turn brown	2
- Identifies bromine water can be used to detect ethene OR - States colour change occurs in presence of ethene with bromine	1

Question 25

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the relationship between DO and BOD and the importance of monitoring - Describes what DO and BOD are and how they are related - Gives reasons for monitoring both - Outlines effect of BOD on water quality - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	5
- Demonstrates sound knowledge and understanding of the relationship between DO and BOD and the importance of monitoring - Outlines DO and BOD with link between the two - Gives reasons for monitoring both - Outlines effect of BOD on water quality - Communicates some scientific principles and ideas in a clear manner	4
- Demonstrates a basic knowledge and understanding of DO and BOD - Outlines DO and BOD – link between two - Outlines that BOD is a measure of water quality - Communicates some scientific principles and ideas in a clear manner	3
- Defines DO or BOD - States importance of DO or BOD - Relates BOD or DO to water quality - Communicates ideas in a basic form using general scientific terms	2
- Defines DO or BOD OR - Relates level of DO or BOD to water quality - Communicates simple ideas	1

Question 26

Criteria	Marks
- Gives correctly balanced equation with correct states of matter - Calculates volume of gas produced	4
- Gives correctly balanced equation - Calculates correct moles of H_2 produced OR - Calculates volume of $H_2(g)$ with correctly balanced equation, using correct mole ratio OR - Provides correct equation without recognising the limiting reagent	3
- Gives correctly balanced equation and correct number of moles of either reactant OR - Gives correct number of both reactants OR - Gives correct number of moles of either and correct volume calculation OR - Gives incorrect equation and moles but gives volume, mole ratio and limiting reagent	2
- Writes correct balanced equation OR - Calculates moles of either reactant	1

Question 27

Criteria	Marks
Correctly identifies all THREE components	2
- Correctly identifies anode and cathode OR - Correctly identifies electrolyte and identifies anode and cathode but in reverse	1

Question 28

Criteria	Marks
- Demonstrates a thorough understanding of all THREE steps by describing features of steps with appropriate reference to techniques and equipment used - Determines concentration of HCl correctly with correct equation	7–8
- Demonstrates a sound understanding of all THREE steps by outlining features of steps with appropriate reference to some techniques and equipment used - Provides ONE correct calculation and correct equation	5–6
- Demonstrates a limited understanding of all THREE steps. Refers to some techniques and equipment used - Calculates moles or mass of Na_2CO_3 or correct equation	3–4
- Demonstrates a basic understanding of some steps or refers to some techniques or equipment used OR - Calculates formula mass of Na_2CO_3 or titrated moles or mass of Na_2CO_3 with some outline of procedure OR - Calculates correct concentration of HCl OR - Provides correct equation and refers to some techniques or equipment used	2

Question 29 (a)

Criteria	Marks
- Lists at least TWO assumptions correctly - Explains validity of experiment in terms of correct outcomes for both	3
- Lists at least TWO correct assumptions - Explains validity of ONE, or outlines validity for both OR - Lists at least THREE assumptions, no assessment	2
Lists ONE or TWO correct assumptions	1

Question 29 (b)

Criteria	Marks
- Gives correctly balanced (must have $\text{BaSO}_4 (s)$) equation - Correctly calculates mass of barium sulfate - Recognises moles of $\text{SO}_4^{2-} = \text{moles of BaSO}_4$	3
- Calculates mass of barium sulfate (with equation) but makes 1 error OR - Calculates mass of barium sulfate correctly without equation	2
- Writes correct equation OR - Correctly calculates mass of sulfate	1

Question 30 (a)

Criteria	Marks
- Identifies both types of polymerisation correctly - Describes each process - Writes TWO correct equations	3
- Identifies both types of polymerisation AND either - Gives TWO correct equations OR - Describes ONE process with correct equation	2
- Identifies both types of polymerisation correctly OR - Outlines main features of each process OR - Gives ONE correct equation OR - Names and outlines process for ONE monomer	1

Question 30 (b)

Criteria	Marks
- Identifies THREE polymers and relates properties of each to structure and uses - Demonstrates coherence and logical progression of ideas	5
- Identifies THREE polymers and relates properties of TWO to structure and uses - Demonstrates coherence and logical progression of ideas	4
- Identifies THREE polymers and relates uses - Relates structure and property of ONE OR - Identifies TWO polymers and relates properties of each to structure and use	3
- Identifies THREE polymers OR - Identifies TWO polymers and related uses OR - Identifies ONE polymer and relates structure to properties and uses	2
- Identifies TWO polymers OR - Identifies ONE polymer and related use	1

Question 31 (a) (i)

Criteria	Marks
Calculates correct percentage of TDS to 2 sig. fig	2
- Calculates correct mass of TDS OR %TDS to incorrect significant figures	1

Question 31 (a) (ii)

Criteria	Marks
Gives a correct chemical test for Cl^- and a balanced chemical equation	2
- Identifies a way to detect Cl^- in a water sample OR - Gives an equation for the detection of Cl^-	1

Question 31 (b)

Criteria	Marks
Correctly identifies a source and effect of a named ion that pollutes waterways	2
- Identifies a source of a named ion contamination OR - Identifies an effect of a named ion being present in waterways	1

Section II

Question 32 – Industrial Chemistry

Question 32 (a)

Criteria	Marks
Identifies type of cell and sketches in general terms the process used to extract NaOH	3
- Identifies the cell AND ONE aspect of process OR - Sketches in general terms the process used to extract NaOH	2
- Identifies the type of cell OR - Identifies a step in the process of extracting NaOH	1

Question 32 (b)

Criteria	Marks
- Thoroughly compares the electrolysis of molten and aqueous sodium chloride and includes relevant half and full equations - Demonstrates coherence and logical progression of ideas and includes correct use of scientific principles and ideas	5
- Presents more than one comparison with half equations and overall reaction for one cell OR - Provides all half equations and full reactions for both types of electrolysis with only one stated comparison	4
- Presents only ONE comparison with half equations and overall reaction for one cell OR - Provides all half equations and full reactions for both types of electrolysis OR - Thoroughly compares the electrolysis of molten and aqueous sodium chloride with no equations	3
- Provides ONE comparison between the electrolysis of molten and liquid NaCl and $\frac{1}{2}$ equations for ONE electrolysis OR - Provides all half equations and full reaction for ONE type of electrolysis OR - Provides ONE comparison between the electrolysis of molten and aqueous NaCl and ONE full reaction OR - Soundly compares the electrolysis of molten and aqueous sodium chloride with no equation	2
- States ONE feature of either electrolysis OR - Writes ONE overall reaction for either electrolysis or TWO $\frac{1}{2}$ equations	1

Question 32 (c) (i)

Criteria	Marks
Provides correct equation constant expression and value with evidence of working out	3
- Provides the correct equilibrium constant expression and concentration for two of SO_2 and SO_3 or O_2 OR - Provides correct chemical equilibrium reaction and equilibrium constant expression OR - Provides correct chemical equilibrium reaction and concentration for two of SO_2 and SO_3 or O_2 OR - Provides correct equilibrium constant expression and moles for SO_2, SO_3 and O_2	2
- Provides the correct equilibrium constant expression OR - Provides correct chemical equilibrium reaction OR - Provides correct concentration for any two of SO_2 and SO_3 or O_2 OR - Provides correct moles for all of SO_2 and SO_3 or O_2	1

Question 32 (c) (ii)

Criteria	Marks
States the condition responsible for the change with an appropriate justification	2
Identifies that temperature was a factor in changing equilibrium	1

Question 32 (d) (i)

Criteria	Marks
Identifies type of reaction and reactant A	2
- Identifies type of reaction OR - Identifies reactant A	1

Question 32 (d) (ii)

Criteria	Marks
Demonstrates a thorough knowledge of how saponification could be carried out in a school laboratory AND identifies at least TWO safety precautions for this activity	3
- Demonstrates a sound knowledge of how saponification could be carried out in a school laboratory - Identifies at least ONE safety precaution for this activity	2
- Demonstrates a limited knowledge of how saponification could be carried out in a school laboratory OR - Identifies at least ONE safety precaution for this activity	1

Question 32 (e)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of the importance of the use of limestone and the resulting environmental impact - Provides relevant equations - Provides a judgement on importance and environmental impact - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of the importance of the use of limestone and the resulting environmental impact - Provides a relevant equation - Communicates some scientific principles and ideas in a clear manner	4–5
- Demonstrates basic knowledge of the importance of the use of limestone and its environmental impact - Communicates ideas in a basic form using general scientific terms	2–3
- Demonstrates a limited knowledge of the importance of the use of limestone and its environmental impact - Communicates simple ideas	1

Question 33 – Shipwrecks, Corrosion and Conservation

Question 33 (a)

Criteria	Marks
Demonstrates a sound knowledge and understanding of the effect the marine environment has on materials	3
Demonstrates a sound knowledge and understanding of the effect the marine environment has on ONE material	2
- Demonstrates limited knowledge of the effect of the marine environment on TWO materials OR - Demonstrates basic knowledge of the effect of the marine environment on ONE material	1

Question 33 (b) (i)

Criteria	Marks
- Draws fully labelled diagram - Provides both relevant half equations - Provides overall reaction equation	4
- Draws labelled diagram with some errors - Provides relevant half equation and full overall equation OR - Draws fully labelled diagram - Provides a relevant half equation	3
- Draws a diagram with some correct labels - Writes ONE correct half equation OR - Writes both relevant half equations and overall equations, no diagram OR - Draws a fully labelled diagram – no equations	2
- Draws a diagram with some correct labels OR - Writes ONE correct half equation	1

Question 33 (b) (ii)

Criteria	Marks
Clearly indicates how the cathode would be identified	1

Question 33 (c)

Criteria	Marks
- Provides a thorough explanation of the properties and related uses of the steels based on the carbon content AND/OR the presence of other elements - Communicates ideas using scientific principles in a logical manner with the reference to the information in the table	5
- Provides a sound explanation of the properties and related uses of the steels based on the carbon content AND/OR the presence of other elements - Communicates ideas using scientific principles with reference to the information in the table	4
Provides a sound understanding of the properties and uses of steels	3
Demonstrates a basic knowledge of the properties and uses of some steels	2
Demonstrates a limited knowledge of the properties OR uses of steels	1

Question 33 (d) (i)

Criteria	Marks
- Provides features of a test which identifies the THREE factors and describes appropriate methods to assess their effect on the rate of corrosion - Demonstrates coherence and logical progression and includes use of scientific principles and ideas	4
- Provides features of a test which identifies THREE factors - Describes relevant tests for TWO factors - Communicates some scientific principles and ideas in a clear manner	3
- Outlines a test for ONE factor, linking the test to the factor - Communicates simple ideas OR	2
Provides features of a test which identifies THREE factors OR	1
Demonstrates a limited knowledge of factors and tests associated with corrosion of iron OR	
Communicates simple ideas	

Question 33 (d) (ii)

Criteria	Marks
Lists ONE environmental relevant method of protection	1

Question 33 (e)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of restoration and conservation techniques for the TWO materials - Compares process for the TWO materials - Uses equations to describe restoration of copper or wood - Provides a judgement on suitability of techniques - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of restoration and preservation of the TWO materials - Provides a judgement on suitability of techniques - Communicates scientific principles clearly and uses an equation when discussing copper or wood restoration	4–5
- Demonstrates basic knowledge and understanding of restoration and preservation of the TWO materials - Communicates in basic form using general scientific principles OR - Demonstrates a sound knowledge and understanding for ONE material - Communicates in basic form using general scientific principles	2–3
- Demonstrates a limited knowledge of EITHER restoration OR preservation for ONE of the materials - Communicates simple ideas	1

Question 34 (a)

Criteria	Marks
- Correctly names molecule - Relates ADP to ATP - States importance in energy transfer and storage	3
- Correctly names molecule - Relates ADP to ATP OR - States importance in energy transfer and storage	2
- Correctly names molecule OR - Links molecule to energy transfer or storage OR - Relates ADP to ATP	1

Question 34 (b) (i)

Criteria	Marks
- States that oxidation of glucose is a multistep process - States that small amounts of energy are produced at each step	2
ONE of the above	1

Question 34 (b) (ii)

Criteria	Marks
- Identifies lactic acid as the cause of change in pH - Recognises pH is lowered - Gives a correctly balanced equation	3
- Identifies lactic acid as the cause of change AND EITHER - Recognises pH is lowered OR - Gives a correctly balanced equation	2
- Identifies lactic acid as the cause of change in pH OR - Recognises pH is lowered OR - Gives a correctly balanced equation	1

Question 34 (c)

Criteria	Marks
Demonstrates a thorough knowledge of the chemical bonding that determines the secondary and tertiary structures of a protein	4–5
- States side chains determine 3-dimensional structure - Outlines THREE of the types of forces - Refers to structure	3
- Outlines TWO types of the forces with reference to structures OR - Outlines THREE types of forces – no reference to structures OR - Lists FOUR forces – no reference to structures	2
- Lists TWO of the forces OR - Outlines ONE force	1

Question 34 (d)

Criteria	Marks
- Identifies THREE stages of aerobic respiration - States correct ATP output at each stage - Identifies TWO intermediates - Constructs a logical and sequential flowchart	5
- Identifies THREE stages of aerobic respiration - Identifies TWO intermediates - States overall ATP but not at each stage - Constructs a logical and sequential flowchart	4
- Identifies THREE stages of aerobic respiration - States overall ATP production - Constructs a simplified flowchart	3
- Identifies TWO stages of aerobic respiration OR - States ATP production and identifies ONE stage OR - Constructs a simplified flowchart	2
- Identifies ONE stage OR - States overall ATP production OR - Constructs simple flowchart	1

Question 34 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of skeletal muscles and how this has influenced training regimes • Justifies use of specific training regimes • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates a sound knowledge and understanding of skeletal muscles – includes both Type 1 and Type 2 cells • States use of specific training regimes • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of skeletal muscles – Type 1 and Type 2 cells • Communicates ideas in basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of skeletal muscles • Communicates simple ideas	1

Question 35 (a)

Criteria	Marks
• Identifies the element • Provides a valid justification for the choice	3
• Identifies the element and the trend across a period OR • Identifies an element with the highest electronegativity in its period but not period 3 and trend across a period OR • Gives a valid explanation for the trend in electronegativity across a period but incorrectly identifies the most-electronegative element	2
• Identifies the element correctly OR • Correctly identifies the trend across a period in the periodic table OR • Defines electronegativity	1

Question 35 (b) (i)

Criteria	Marks
• Identifies correct ligand type	1

Question 35 (b) (ii)

Criteria	Marks
- States colours correctly - Identifies the number of valence electrons correctly and relates it to energy levels - Valid explanation of the presence or lack of colour for the two complex ions	4
- States colours correctly AND EITHER - Identifies the number of valence electrons correctly OR - Gives a valid explanation of the presence or lack of colour for the two complex ions	3
- States colours correctly - Gives a limited explanation of the presence or lack of colours OR - Explains the basis of colour	2
- States the colour for ONE complex ion OR - Number of valence electrons for ONE OR - Identifies reasons for colour	1

Question 35 (c) (i)

Criteria	Marks
- Draws correct reflectance spectrum - Identifies pigment colour	2
- Draws resulting reflectance spectrum OR - Identifies pigment colour	1

Question 35 (c) (ii)

Criteria	Marks
• Outlines IR radiation • Outlines method of use and target compounds/pigments • Outlines advantages and disadvantages of the method of analysis	3
• Outlines method of use and target compounds/pigments AND EITHER • Outlines advantages and disadvantages of the method of analysis - OR • Outlines IR radiation	2
• Identifies TWO or more pigments/compounds detected by IR - OR • Outlines the method of analysis with advantages and risks - OR • Outlines the advantage or disadvantage of the method of analysis	1

Question 35 (d)

Criteria	Marks
- Provides a thorough discussion of the Bohr model by: - Describing the Bohr model of the atom - Applying the Bohr model to explain phenomena of emission spectra - Stating a merit of the Bohr model was that it predicted spectrum for hydrogen - Referring to spectra of other elements to demonstrate at least two limitations - Demonstrates coherence and logical progression of ideas using scientific principles	4–5
- Provides a thorough discussion of the Bohr model by: - Outlining the Bohr model of the atom - Applying the model to explain the phenomena of emission spectra - Stating a merit of the Bohr model - Referring to spectra of other elements to demonstrate a limitation	2–3
- Outlines the Bohr model of the atom OR - Applies the Bohr model to explain the phenomena of emission spectra OR - States a merit of the Bohr model OR - Refers to spectra of other elements to demonstrate a limitation of the model	1

Question 35 (e)

Criteria	Marks
• Demonstrates thorough knowledge and understanding of potential health risks associated with cosmetics used by ancient cultures • Provides a judgement on use of these cosmetics over time • Demonstrates coherence and logical progression and includes TWO pigments, with chemical formulae and names	6–7
• Demonstrates sound knowledge and understanding of potential health risks associated with cosmetics used by ancient cultures • Identifies ONE pigment with chemical formula and name • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of potential health risks of cosmetics of ancient cultures • Identifies ONE pigment with its use and risk • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of the potential health risks of cosmetics of ancient cultures OR • Identifies ONE pigment • Communicates simple ideas	1

Question 36 (a)

Criteria	Marks
• Identifies A and B • Sketches in general terms the difference in structure • Identifies the origin of A and B	3
TWO of: • Identifies A and B • Sketches in general terms the difference in structure of named A and B • Identifies the origin of named A and B	2
• Identifies A and B OR • Sketches in general terms the difference in structure of named A and B OR • Identifies the origin of named A and B	1

Question 36 (b) (i)

Criteria	Marks
- Identifies that the amino acids move because of an electrical potential difference - Identifies factors that cause the separation of amino acids	2
- Identifies that the amino acids move because of an electrical potential difference OR - Identifies factors that cause the separation of amino acids	1

Question 36 (b) (ii)

Criteria	Marks
- Identifies each amino acid correctly with an appropriate reason - Clearly indicates that the top of the tray is positive (or the bottom is negative)	3
Gives a correct identification and appropriate reason for two amino acids	2
- Gives a correct identification and appropriate reason for one amino acid OR - Identifies all three amino acids	1

Question 36 (c) (i)

Criteria	Marks
Identifies the manufacturer of the pottery	1

Question 36 (c) (ii)

Criteria	Marks
Demonstrates thorough knowledge and understanding of emission spectra	4
Demonstrates sound knowledge and understanding of emission spectra	3
Demonstrates a basic knowledge of emission spectra	2
Demonstrates a limited knowledge of emission spectra	1

Question 36 (d) (i)

Criteria	Marks
Names FOUR appropriate properties	2
Names TWO appropriate properties	1

Question 36 (d) (ii)

Criteria	Marks
- Names an appropriate organic and an inorganic test - Gives description of both tests	3
- Names an appropriate organic and an inorganic test - Gives a description of ONE test	2
- Names an appropriate organic and an inorganic test OR - Gives a description of ONE test	1

Question 36 (e)

Criteria	Marks
- Demonstrates thorough knowledge and understanding of both the polymerase chain reaction and DNA electrophoresis and their application in forensic analysis - Provides a range of uses for DNA analysis - Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
- Demonstrates sound knowledge and understanding of the process used to analyse DNA - Provides some uses for DNA analysis - Communicates some scientific principles and ideas in a clear manner	4–5
- Demonstrates basic knowledge of DNA analysis - Provides a use for DNA analysis - Communicates ideas in a basic form using general scientific terms	2–3
- Demonstrates a limited knowledge of some of the processes of DNA analysis OR - The use that this analysis can be put to - Communicates simple ideas	1

Chemistry

2010 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.2.2.3	H9
2	1	9.2.5.2.2	H6
3	1	9.3.1.2.2	H8
4	1	9.4.4.2.9	H9
5	1	9.4.5.3.2, 9.4.5.2.1	H8
6	1	9.2.1.2.7	H9
7	1	9.3.3.2.2, 9.3.3.2.6	H10
8	1	9.3.4.2.1	H2
9	1	9.4.2.2.2	H8
10	1	9.4.3.3.1, 9.4.5.2.1, 9.4.3.2.1, 14.1d	H8, H14
11	1	9.2.3.2.1, 9.2.3.2.9	H9
12	1	9.2.4.2.3, 12.4b	H10, H12
13	1	9.2.4.2.4, 9.2.4.2.6	H8
14	1	14.1a	H9, H14
15	1	9.2.3.3.4, 9.2.3.3.5, 14.1d	H12, H10
16	1	9.4.4.2.5, 13.1e	H6
17	1	12.4b, 14.1a	H12
18	1	9.3.2.2.4	H8
19	1	9.3.2.2.9, 12.4b	H10, H12
20	1	12.3c, 12.4b, 14.1a, 14.1f	H10, H12, H14
Section I			
Part B			
21	3	9.3.3.2.3, 9.3.3.2.4, 9.3.3.2.6, 9.3.3.2.7, 14.1g	H8, H14
22 (a)	1	9.3.5.3.2	H9
22 (b)	2	9.3.5.2.6, 9.3.5.3.1	H8, H9
22 (c)	3	9.3.5.2.2, 13.1e	H9, H10, H13
23 (a)	1	9.2.3.3.6	H9, H10
23 (b)	2	9.2.3.3.6, 12.4b	H9, H10, H12
24 (a)	2	9.2.1.2.3	H10, H9
24 (b)	2	9.2.1.3.2	H9
25	5	9.4.5.2.1, 14.3b	H4, H8, H14
26	4	9.2.1.3.1, 9.3.2.2.9, 12.4b	H8, H10, H12
27	2	9.2.4.2.4, 9.2.4.2.6	H8, H10
28	8	9.3.4.2.8, 9.3.4.3.3, 12.4b, 11.3a	H8, H10, H11, H12
29 (a)	3	9.4.3.3.3, 11.2e, 12.4a	H8, H10, H11, H12
29 (b)	3	9.4.3.3.3, 12.4b	H8, H10, H12

Question	Marks	Content	Syllabus outcomes
30 (a)	3	9.2.1.2.5, 9.2.1.2.6, 14.1f, 9.2.2.2.3, 9.2.2.2.2, 9.2.2.2.4, 14.3d	H9, H10, H14
30 (b)	5	9.2.1.2.8, 9.2.2.4, 9.2.2.2.5, 14.3d	H4, H9, H14
31 (a) (i)	2	9.4.5.2.1, 12.4d	H12
31 (a) (ii)	2	9.4.3.3.1, 11.2c	H8, H10, H11
31 (b)	2	9.4.3.3.2	H4

Section II

Question 32 — Industrial Chemistry

32 (a)	3	9.5.4.2.2, 9.5.4.2.3, 12.3c	H7, H8, H12
32 (b)	5	9.5.4.3.1, 9.5.4.3.2, 14.3b	H13, H7, H14
32 (c) (i)	3	9.5.2.2.2, 9.5.2.3.3, 12.3c	H12, H13
32 (c) (ii)	2	9.5.2.2.1, 9.5.2.2.3, 12.3c, 14.1d	H12, H14
32 (d) (i)	2	9.5.5.2.1, 12.3c	H9, H10, H12
32 (d) (ii)	3	9.5.5.2.1, 9.5.5.2.2, 9.5.5.3.1, 12.1d	H11, H12
32 (e)	7	9.5.6.2.1, 9.5.6.2.3, 9.5.6.2.4, 14.3b, 13.1d	H4, H13, H14

Section II

Question 33 — Shipwrecks, Corrosion and Conservation

33 (a)	3	9.6.7.2.1, 9.6.2.2.4, 14.1d	H8, H14
33 (b) (i)	4	9.6.3.2.1, 9.6.3.2.2, 13.1d	H7, H8, H10, H13
33 (b) (ii)	1	9.6.3.2.1, 14.1d	H7, H14
33 (c)	5	9.6.2.2.3, 9.6.2.3.3, 14.1c, 12.3c	H8, H12, H14
33 (d) (i)	4	9.6.5.2.1, 9.6.5.3.1, 11.3a, 12.2b	H4, H6, H7, H8, H11, H12
33 (d) (ii)	1	9.6.4.2.1, 14.2a	H4, H14
33 (e)	7	9.6.7.2.3, 9.6.7.2.4, 9.6.7.2.5, 13.1d 14.1g, 14.2a, 14.3b	H3, H13, H8, H14

Section II

Question 34 — The Biochemistry of movement

34 (a)	3	9.7.1.2.3, 9.7.1.3.1, 14.1f	H7, H14
34 (b) (i)	2	9.7.8.2.2, 9.7.8.2.5, 9.7.8.3.2, 9.7.8.2.3, 14.1g	H7, H8, H14
34 (b) (ii)	3	9.7.10.2.2, 13.1 d, 14.1a	H9, H10, H12, H13
34 (c)	5	9.7.4.2.1, 9.7.4.2.2, 9.7.4.2.5, 14.1f	H9, H14
34 (d)	5	9.7.8.3.1, 13.1e	H7, H9, H10, H13
34 (e)	7	9.7.10.2.1, 9.7.8, 9.7.10, 14.3b	H1, H3, H7, H14

Section II

Question 35 — The Chemistry of Art

35 (a)	3	9.8.3.2.7, 14.1a	H6, H14
35 (b) (i)	1	9.8.5.2.5, 114.1f	H14
35 (b) (ii)	4	9.8.4.2.4, 14.1d	H14
35 (c) (i)	2	9.8.2.2.4, 12.3c, 13.1f	H12, H13
35 (c) (ii)	3	9.8.2.2.8, 9.8.2.3.3, 14.2d	H14
35 (d)	5	9.8.2.3.4, 9.8.2.2.3, 9.8.2.2.5, 14.1c	H1, H2, H6, H14
35 (e)	7	9.8.1.3.2, 9.8.1.2.3, 14.1b	H1, H4, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 36 — Forensic Chemistry			
36 (a)	3	9.2.2.2.2, 9.9.2.2.4, 9.9.2.3.2, 13.1e	H9, H13
36 (b) (i)	2	9.9.3.2.2, 9.9.3.2.6, 9.9.3.2.3, 9.9.3.3.5, 11.2c	H8, H9, H11
36 (b) (ii)	3	9.9.3.2.2, 9.9.3.2.6, 9.9.3.2.3, 9.9.3.3.5, 13.1e	H8, H9, H13
36 (c) (i)	1	9.9.6.2.2, 9.9.6.2.5, 9.9.6.2.3, 9.9.6.3.2, 14.1a	H7, H14
36 (c) (ii)	4	9.9.6.2.1, 9.9.6.2.3, 9.9.6.2.2, 9.9.6.2.4, 14.1f	H7, H14
36 (d) (i)	2	9.9.1.2.4, 9.9.1.3.3, 11.2c, 12.1a	H8, H12
36 (d) (ii)	3	9.9.1.1.1, 9.9.1.3.3, 9.9.1.2.4, 11.2c, 12.2a	H4, H8, H11, H12
36 (e)	7	9.9.4.2.2, 9.9.4.3.1, 9.9.4.2.3, 14.3b	H4, H9, H14